Student Dietary Habits — Project Overview

A Tableau-driven analysis of 120 students that links eating habits, exercise, and emotional drivers to weight and academic performance. The work transforms raw survey data into actionable insights for university wellness programs, administrators, and health coordinators.

Problem Statement

Academic pressure, irregular schedules, and stress lead many students to skip meals or choose comfort foods. These behaviors can harm physical and mental health, and may influence GPA and overall quality of life. This project diagnoses those patterns to inform targeted wellness interventions.

Objectives

Consumption Patterns

Quantify breakfast choices, frequency of eating out, and cooking habits.

Diet ⇄ Academic Performance

Analyze relationships between nutritional habits and GPA distribution.

Exercise & Weight

Assess how exercise frequency correlates with weight ranges.

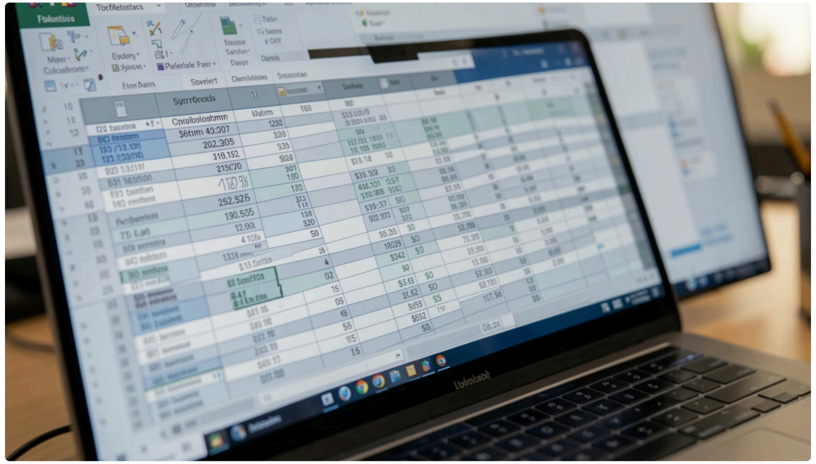
Emotional Eating

Identify leading comfort-food drivers (stress, boredom, sadness).

Tableau Dashboard

Build an interactive dashboard to surface KPIs and support decisions.

Data & Preparation



Source: food_coded.csv — 120 student survey records. Key attributes: Gender, GPA, Grade Level, Weight, Breakfast Preference, Eating Out Frequency, Cooking Frequency, Exercise Level, Health Category, Comfort Food Reasons, Favorite Foods, Nutritional Habits.

Cleaning steps: missing-value checks, duplicate removal, data-type correction (GPA → decimal, Weight → numeric), category standardization (e.g., Male/Male → Male), and final quality verification before Tableau import.

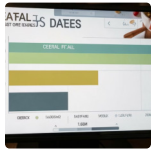
Tableau Import & Dashboard Architecture

Connection: CSV → Tableau Desktop. Actions taken: set dimensions/measures, assign data types, create calculated fields, and structure the workbook into worksheets feeding a single interactive dashboard with KPI cards, comparative charts, and drilldowns for filters (gender, grade level, exercise level).

Key Visuals — What the Dashboard Shows

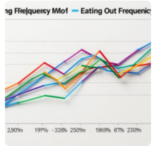
KPI Summary

Total students: 120 • Avg GPA: 3.4 • Avg weight: 158.5 lbs • Avg grade level: 2.4



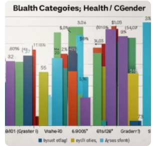
Breakfast Preferences

Cereal is the most common breakfast; donuts are rare—implication: many students have an accessible, healthier option.



Cooking vs Eating Out

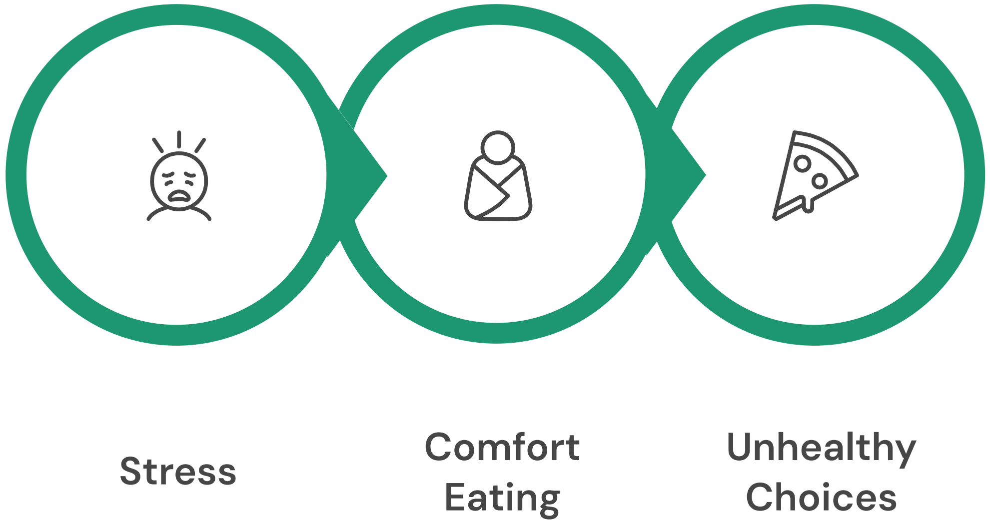
Inverse relationship: students who cook more tend to eat out less, suggesting cooking promotion could reduce outside consumption.



Health Category by Gender

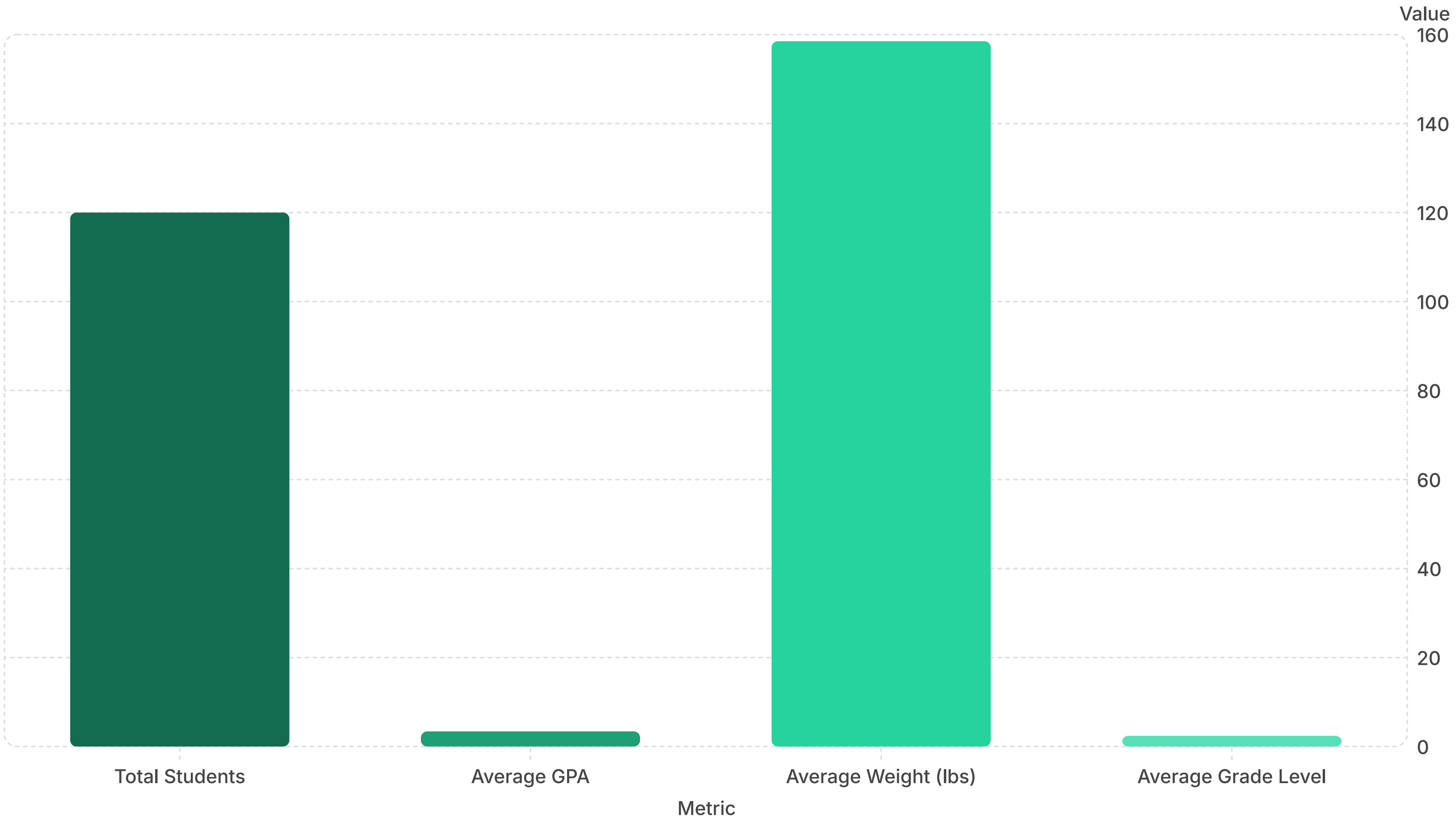
Females appear more often in the healthy category; unhealthy behavior exists across genders.

Findings — Behavioral & Emotional Drivers



The diagram summarizes the primary pathway observed: stress → comfort eating → poorer dietary choices. Tableau bubble charts show stress as the dominant reason for comfort-food consumption, followed by boredom and sadness.

Quantitative Insights



These headline numbers anchor the dashboard. Use them as quick reference KPIs for health-program benchmarking and to segment further analysis by cohort.

Recommendations for University Wellness Programs

Promote Accessible Breakfasts

Partner with dining services to ensure affordable cereal, fruit, and whole-grain options during peak morning hours.

Cooking Skills Campaign

Offer short workshops and quick-recipe kits to increase cooking frequency and reduce reliance on eating out.

Integrate Mental Health Supports

Target stress reduction (counseling, peer groups, mindfulness) to reduce emotional eating drivers.

Active Lifestyle Incentives

Design exercise challenges and integrate wearables to encourage consistent activity linked to healthier weight profiles.

Next Steps & Future Scope

Short-term: expand sample size, add semester-level tracking, and publish the Tableau workbook to a web portal for stakeholder access.

Medium-term: integrate wearable and meal-tracking data; pilot personalized diet nudges informed by dashboard signals.

Long-term: apply predictive models to identify at-risk students and tailor preventive interventions.

This dashboard empowers administrators and health coordinators to design targeted, data-driven student wellness initiatives grounded in real behavioral patterns.